A-A-50219
20 January 1984

COMMERCIAL ITEM DESCRIPTION

WHEEL LATHE

RAILWAY, DUAL END DRIVE

The General Services Administration has authorized the use of this commercial item description.

This Commercial Item Description covers the requirements for wheel lathe for profiling the treads of mounted wheel sets with or without mounted roller bearings, traction motors and gears.

Salient characteristics:

Self-aligning headstock.
Infinitely variable spindle-speed from 2.25 to 24 rpm.
Infinitely variable feed rates.
Maindrive motor or motors shall be variable speed AC, 50 HP minimum.
Accept wheel tread diameters from 26 to 52 inches.
Two sets of compound tools slide with hydraulic feed and tracer system.
Capability to trace wide and narrow flanges.
Provided with hydraulic or spring-loaded centers.
Provided with hydraulic pressurized jaws.
Speed indicator to indicate actual RPM of machine in operation.
Digital component two axial, 0.001 in resolution, numbers of 3/4 inch minimum height with induction pick-up.
Chuck jaws to have insert type changeable bits.
Chuck jaws shall be adjustable individually both laterally and longitudinally.
Hydraulic governor valve with control knob to vary feed rate.
Four position toggle switch to control direction of feed and rapid transverse.
Provided with forced lubrication.
Carbide steel bit inserts or equivalent.
Wheel profiles templates.
Painting and finishing of the machine shall be the manufacturer's standard commercial practice.
Permanent metal identification plate mounted in a conspicuous place and showing Manufacturer's Name, Address, Contract Number, Model, Serial Number, Date of Manufacture, and Power Input Characteristics and Rating.

Design and construction. All components shall be designed to form an integral part of a compact unit conforming to the. requirements listed herein. The unit shall be designed with ready access to components for adjustments, replacement, or repair, and to facilitate set-up, operation and maintenance without the need for special tools. MIL-STD-14721, 5.9 requirements apply to design

FSC 4940

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for maintainability. Special tools that are required shall be provided by the contractor. The machine shall be complete so that when installed and connected to the power source, it can be used for any operation for which it was designed. Construction shall be such that the unit will retain its performance and accuracy, and function properly when operated at its maximum functional capacity.

Materials and workmanship. Materials shall meet ASTM specifications where applicable for the purpose intended. Workmanship in construction and finish shall be of high quality and consistent with industry standards. The machine shall be constructed of parts which are sound in structure and free from any repair. All parts shall be new and unused. Castings, forgings, molded parts, stampings, and welded parts shall be cleaned and free from sand, dirt, fins, sprues, scale flux, and other harmful or extraneous materials. External surfaces shall be smooth and all edges shall be either rounded or beveled, except where a sharp edge is required by design.

Essential safety features. The machine shall be fitted with suitable devices and covering or guards to provide protection from electrical shock and moving parts such as gears, belts, pulleys, and other parts which may present a hazard. All guards shall be designed to provide ready access to the guarded parts. All controls shall be within convenient reach and vision of operator in his normal operating position. The lathe shall meet all applicable OSHA requirements. The machine shall be provided with a prominently located warning sign which cautions the operator about unsafe operating practices. MIL-STD-1474, 4.3 and 5.1 shall apply.

Lubrication requirements. All bearings (except sealed-for-life type), mating gears, and sliding parts shall be provided with a means for lubrication. All oil holes, grease fittings, and filler caps shall be easily accessible. A lubrication chart or plate shall be securely attached to the unit. The chart shall be made of black anodized aluminum with white nomenclature/letters or other similarly durable material and shall be permanently attached to the unit, lettering in accordance with MIL-STD-1472, 5.5. The chart or plate shall provide the following information.

POINTS OF LUBRICATION
SERVICE INTERVAL
TYPE OF LUBRICATION
VISCOSITY

Standard Commercial Product. The above salient characteristics are set forth as minimum requirements. The lathe shall be in accordance with these minimum requirements and shall be the manufacturer's standard commercial product. Additional or better features which are not ' specifically prohibited by this description but which are a part of the manufacturer's standard commercial product, shall be included in the lathe being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements., manufacturer's catalogs, or brochures, and represents the latest production model. It shall have been in production, marketed, and in use for a minimum of one year. The introduction of normal product improvements will be considered by the procuring agency upon

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written request of the vendor. The procuring agency reserves the right to grant or not to grant incorporation of such improvements. The bidder shall submit proof of this offering as a commercial product with his bid. The contracting officer will determine the acceptability of this proof. Failure of the bidder to furnish proof of his offering as his standard commercial product shall be cause for rejection of his bid.

Acceptance. Acceptance of each lathe shall be based on visual inspection and demonstration of its performance capabilities. Unless otherwise specified, the contractor is responsible for the performance of all inspection requirements. The Government shall completely inspect the lathe for design, construction, materials, components, electrical equipment and workmanship to determine compliance with these requirements. The lathe shall be given at least a 1 hour continuous run test with or without work at the Government's discretion, combining all operations and all speed ranges up to maximum speed permissible. Any deviation from these requirements shall be cause for rejection.

Packaging and marking. Military packaging shall be specified when shipments are to overseas Department of Defense (DOD) consignees. Commercial packaging may be specified for shipments to DOD consignees within the Continental United States and to non-DOD consignees worldwide. When a supplier's normal packaging will satisfy the requirements for military packaging, it may be used.

Military packaging. Each wheel lathe shall be preserved and packed to protect against physical damage and deterioration in a logistic environment characterized by the following:

1. Multiple handling during transportation and intransit storage.
2. Shock, vibration, and static loading forces during worldwide shipment by all transport modes.
3. Uncontrolled warehouse storage for extended periods.
4. Weather exposure during shipment and intransit transfers.
5. Stacking loads during shipment and storage.

Commercial. Each wheel lathe shall be packaged in accordance w1th the supplier's normal method for retail distribution. Shipping containers shall comply with Uniform Freight Classification or National Motor Freight Classification rules.

Marking. Unit, intermediate (when provided), and exterior packs shall be marked with the item nomenclature, stock or part number, quantity per pack, and contract number as a minimum; additional marking shall be as specified in the contract order. Exterior packs shall have in addition the complete consignee's address. Markings shall be applied in a legible manner, in a contrasting color, or an appropriate size, and of waterproof materials.

Custodian

Army – ME

PREPARING ACTIVITY

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Project 4940-A453

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